

UNIT 5

Software Tools and Interactive Devices

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Unit Outcomes

- **Choose between different software tools for building interfaces.**
- **Select the appropriate devices for interaction based on the application.**

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Components of UI Software

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Introduction

- User interface software is most of the time large, complex, and difficult to implement, debug, and modify.
- An average of 48% of the application code is of the user interface, and that about 50% of the time is taken to implement it.
- In order to help design and implement the user interface software, special software systems and tools have been developed.
- These tools increase the productivity of the developers and many of them are commercial products today.

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Components of User Interface Software

The User Interface software is divided into various layers:

- The windowing system
- The toolkit
- Higher-level tools

Applications
Higher - Level Tools
Toolkit
Windowing System
Operating System

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Components of User Interface Software

The Windowing System:

- Helps in the separation of the screen into different rectangular regions, called windows.
- Some operating systems separate windowing systems into two functional layers: Window system and Window manager.
- Window System: It is the functional or programming interface and provides procedures that allow the application to draw pictures on the screen and get input from the user.

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Components of User Interface Software

- **Window manager:** is the user interface and allows the end user to move windows and display the title lines, borders, and icons around the windows.

Toolkit: contains many commonly used widgets such as menus, buttons, scroll bars, and text input fields.

Higher-level tools: Help the designer use the toolkit widgets.

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Software Tool

- A software tool is a set of computer programs that are used by developers to create, maintain, debug, or support other applications and programs.
- It can also be defined as a program that is employed in the development, repair, or enhancement of other programs or of hardware.
- A software tool can include guidelines and methodologies for interface development.

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Commonly Used Software Tools

- **Specification methods:** These are methods used to specify a Graphical User Interface.
- **Interface building tools:** These are design methods that help in designing command languages, data-entry sequences, and widgets.
- **Evaluation tools:** Tools to evaluate the correctness and completeness of the programs.

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Specification Methods

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Specification Methods

- The user interface design process becomes easy if the designer describes these interfaces clearly and precisely.
- There are many formal specification techniques used in software engineering to describe many aspects of software systems.
- But these techniques have not been widely applied to the problem of specifying user interfaces.

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Steps in the design of a User Interface

- The designer describes and studies a variety of interfaces through a formal specification technique.
- The required user interface is chosen and before it is built, its performance is predicted by applying a Human performance model derived from empirical data.
- A prototype or mockup of the user interface can be built directly from the specification using an executable specification language.

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Steps in the design of a User Interface

- This mockup can then be used to gather experimental data about the proposed user interface early in the design process.

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Commonly Used Specification Methods

Grammars:

- Grammars are useful to specify textual commands or expressions that are understandable by a program.
- They provide confirmations for completeness and correctness.
 - These were earlier used with terminal-based interfaces and today are used on interactive systems.
- Grammars are useful to verify the validity of forms filled in by users on a computer. Example: Telephone book entries.

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Commonly Used Specification Methods

Menu-selection and dialog-box trees:

- These are simple structures that guide the designers in creating a selection style for many applications.
- The specification methods use online drawing tools to construct menu trees so that the designers and users can see the entire tree at one time.

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Commonly Used Specification Methods

Transition Diagrams:

- Transition diagram has a set of nodes that represent system states and a set of links between the nodes that represent possible transitions.
- Each link is labeled with the user action that selects that link and possible computer responses.

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Commonly Used Specification Methods

State Charts:

- It is the representation of a state machine that depicts the flow of control from one state to another.
- These are the extensions of conventional state-transition diagrams in terms of the notions of hierarchy, concurrency, and communication.
- These diagrams convert the language of state diagrams into a highly structured and economical description language.

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Interface Building Tools

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Interface Building Tools

Some interface-building tools and their functionalities are as follows:

- **Specification methods:** are used in the design of components of the system, such as command languages, data-entry sequences, and widgets.
- **Screen-transition diagrams drawn or printed on paper:** help in providing an overview of the system.

These two tools allow UI architects, designers, managers, users, and software engineers to discuss and prepare a paper-based design.

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Interface Mockup Tools

- These are tools used to develop a quick sketch of a GUI.
 - **Example:** Visual Studio .Net, etc.
- In the earlier stages of design, user interface architects create quick sketches showing multiple alternatives in the design.
- This will allow communication within the design team, and convey to the clients the look of the product
- Designers can build user-interface prototypes with multimedia construction tools, such as Dreamweaver.

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Interface Mockup Tools

- User interface mockups can be created with paper and pencil, word processors, or slide show presentation software (such as Microsoft PowerPoint or Apple Keynote).

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Software Engineering Tools

- These are programming tools to provide user interface management systems.
- Programmers build user interfaces with general-purpose programming languages such as Java, C#, or C++.
- But due to the lack of uniform terminology used to describe the tools and their features, choosing among them is a complex task.

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The Windowing System Layer

- A windowing system is a component of a graphical user interface (GUI) in a desktop environment.
- It supports the implementation of window managers and provides basic support for graphics hardware, and pointing devices such as mice, and keyboards.
 - Example: The mouse cursor is drawn by the windowing system.
- It implements graphical primitives such as rendering fonts or drawing a line on the screen.

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GUI Toolkit Layer

- GUI toolkits are user interface program libraries that provide the designers with common widgets, such as windows, scroll bars, pull-down or pop-up menus, data-entry fields, buttons, and dialog boxes.
- Sometimes a GUI tool kit library is available with a programming language. Here an experienced programmer can have great flexibility in the interface design.

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GUI Toolkit Layer

- Some toolkits are available without interactive support. These require time for programmers to gain proficiency.
- Example: Microsoft Windows Forms (Winforms) is a cross-platform toolkit, Apple Macintosh Toolkit (SwiftUI), and Unix X Toolkit (Xtk).

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Keyboard and Function Keys

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Background

- Since 1960, there has been steady progress in processor speeds and storage capabilities and to match these the input and output devices also have advanced in their features.
- Earlier ten-character-per-second teletypes have been replaced by high-speed mega-pixel graphical displays.
- Pointing devices like mice and touchscreens have been added to free users from keyboards.

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Background

- Today gestural input, two-handed input, three-dimensional pointing, voice input/output, and wearable devices have been developed to meet the needs of varied users.

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Keyboards and Keypads

- For textual data entry, keyboards are used.
- Keyboard size and packaging can provide different user satisfaction and usability.
- Large keyboards are used in a professional environment whereas a small keyboard is used when a user requires to enter data and manipulate objects simultaneously.
- Very tiny keyboards and touchscreens are used where there is limited text entry such as in mobile phones.

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Keys

Following key features are implemented on today's keyboards:

- The keys have concave surfaces for proper contact.
- The keys are of a matte finish to reduce glare and slipping of the finger.
- The keys use a 40-125 gram force and a displacement of 1 to 4 millimeters for better typing speeds and low error rates.
- When the key is pressed, it gives a very light click giving tactile and audible feedback to the user.

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Keys

- Keys like the space bar, Enter key, Shift key and Ctrl keys should be larger than the others to allow easy and reliable access.
- Other keys such as Caps Lock and Num Lock should indicate the pressed state with an embedded light.
- The placement of the cursor-movement keys should be properly done for rapid and error-free use.

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Function Keys

- There are twelve function keys F1 to F12 on the keyboard.
- These keys are used to perform different functions.
- These key presses cause the operating system to command the interpreter or an application to perform certain actions on the screen.
- On a Laptop, the Fn key must be pressed along with other keys from F1 to F12 for use.

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Pointing Devices

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Pointing Devices

- These devices unlike keyboards, use a pointer to point to certain items to select them.
- These are very convenient during the use of computer-assisted design tools, drawing tools, or air-traffic control systems where complex information is displayed on the screen.
- The users can avoid learning commands and typographic errors on a keyboard are reduced.

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Pointing Devices

- Pointing devices are very useful for small devices and large wall displays.
- Pointing devices can be categorized based on two factors:
 - Diversity of tasks, the variety of devices, and the strategy of using them.
 - Physical attributes like type of movement (linear or rotational), the dimensionality of movement (1,2, 3...), and positioning (relative or absolute).

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Pointing Tasks

Pointing devices are used for the following types of interaction tasks:

- Select: Here users choose from a set of items.
 - Example: Choosing a part in automobile design
- Position: Here users point to an item in a one, two, or three-dimensional space.
 - Example: Dragging and dropping a block of text in a figure.

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Pointing Tasks

- **Orient:** Here users choose a direction in a one, two, or three-dimensional space.
 - **Example:** Rotating a symbol on a screen.
- **Path:** Here the users perform a series of positioning and orientation operations.
 - **Example:** Drawing a curved line.
- **Quantity:** Here users specify a numeric value.
 - **Example:** Setting a parameter like a page number in a document.

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Pointing Tasks

- **Gesture:** Users indicate to perform an action through a simple gesture.
 - **Example:** Swipe motion to turn a page forward or backward.
- **Text:** Here users can enter, move or edit the text in a two-dimensional space.
 - **Example:** Indicating the location of an insertion or deletion.

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Types of Pointing Devices

There are two basic categories of pointing devices:

- **Direct-control:** These devices directly control the screen surface.
 - These are easy to learn and use.
 - Examples: LightPen, Touchscreens, Stylus
- **Indirect-control:** These devices offer indirect control away from the screen surface.
 - These take time to learn and use.
 - Examples: Mouse, Joystick, Trackball

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Speech and Auditory Interfaces

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Introduction

- With the help of speech and auditory interfaces, users can interact with computers with the help of speech.
- This has become possible with the progress in hardware technology.
- But practical applications can be successful only if the cognitive load on the users and error rates are less.

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Use of Auditory Interfaces

- For users with vision impairments
- For hand-free interaction
- When mobility is required
- When user's eyes are occupied
- During a condition when the keyboard cannot be used

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Challenges of designing Speech Recognition Systems

- Cognitive load is more compared to pointing devices
- Interference from environmental noise
- Variation in recognition across different users and time.

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Challenges of designing interfaces for Speech output

- Slow pace of speech output
- Ephemeral nature of speech
- Difficulty in scanning and searching

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Technologies used by Speech Systems

- Discrete-word recognition
- Continuous-speech recognition
- Voice information systems
- Speech generation
- Non-speech auditory interface

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Discrete-Word Recognition

- The devices with discrete-word recognition capability can recognize individual words spoken by a person.
- Accuracy is 90 to 98% for 100 – 10,000-word vocabulary.
- There are two types of discrete-word recognition systems: Speaker dependent and Speaker independent.
- Used to control wheelchairs, operate equipment, or use computers by people with disabilities like paralysis or unable to use their hands due to an injury.

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Continuous-Speech Recognition

- These systems enable users to dictate letters and compose reports verbally for automatic transcription.
- They also enable automatic scanning and retrieval from radio and television programs, lectures, telephone calls for specific words and topics.
- Major challenges for designing such systems are recognizing the boundaries between spoken words, different accents, variable speaking rates, disturbing background noise.

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Voice Information Systems

- These systems are also known as Interactive Voice Response (IVR) and use voice for communication.
- These are commonly used to provide telephone-based information about some government services and tourist sites.
- They provide voice prompts to guide users about flight timings, can help reserve tickets, and so on.
- These systems offer services at a lower cost.
- Example: Audio tours in museums and audio books

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Speech Generation

- It a technology used to produce human speech artificially.
- Speech generation and digitized speech segments are used when messages are simple and short, messages are to be given at a particular time, and require an immediate response.
- This technology has been used to produce inexpensive, compact, and reliable systems for automobile navigation, internet services, utility control rooms, and games.

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Speech Generation

- They are used to give spoken warnings in cockpits and control rooms.
- These systems are advantageous to users when their visual channels are overloaded, they need to move around and when the environment is too bright to see a message on a display.

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Non-Speech Auditory Interfaces

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Introduction

- The non-speech auditory outputs include audio tones and information presented as a combination of sound and music.
- Example: Teletypes in the earlier days used a bell tone to alert the users of an incoming message or the paper had run out.
- Non-speech audio messages can communicate to the computer user without interfering with an application
- Example: Today's computers use a range of tones to indicate warnings or the completion of a task.

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Types of Sounds

There are three types of sounds created for generating different types of outputs:

- **Auditory icons: Familiar sounds**
 - These are used to reinforce the visual experience in a graphical user interface, such as a door opening, liquid pouring or a ball bouncing in games.

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Types of Sounds

- **Earcons: Unfamiliar sounds which are to be learned to understand their meaning.**
 - These are used to draw the attention of the users with a sharp rising tone as used in mobile devices or in a control room.
- **Cartoonified: These are familiar sounds used in a different way to add realism and engage users in a powerful way as in cartoon animations.**

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